# Kun Zhang

PhD Applicant — Large Language Models: Post-Training, Alignment, and Evaluation

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My work focuses on the evaluation, alignment and post-training of large language models.

## Education

**University of Leeds** — Leeds, United Kingdom
*MSc Advanced Computer Science — taught and assessed entirely in English* — Sep 2025 – Sep 2026

- Dissertation: tests whether the text of company filings helps forecast stock volatility once price history and firm identity are taken into account.
- Selected module marks: Deep Learning **84** • Knowledge Representation & Reasoning **81** • Data Mining & Text Analytics **81** • Machine Learning **77**

**Hunan University of Finance and Economics** — Changsha, China
*BSc Computer Science and Technology* — Sep 2021 – Jun 2025

## Publications

All papers below are **first-authored**; five accepted, three under review. Papers marked † were co-authored with my MSc supervisor Dr Chunwei Xia as corresponding author; the remaining two are **sole-authored**. Venue ranks: ICORE2026 (CORE).

### ACCEPTED

**[1] K. Zhang**, C. Xia†. *Same Exam, Different Scores: A Tokenizer-Grounded Audit of Evaluation-Protocol Sensitivity on Chinese Benchmarks.* **NLPCC 2026** (Resources and Evaluation track) — **Oral**, Springer LNCS.

Multiple-choice benchmarks are often scored by which answer label's first token the model rates highest. Under some tokenizers all four circled-digit or full-width labels share one first token, so the rule cannot pick one. From the tokenizers alone, before running any model, I correctly predicted which of 64 model–label-set pairs break this rule. Separately, scoring written-out answers moved rankings far beyond sampling noise, a shift largely consistent with format errors.

**[2] K. Zhang**, C. Xia†. *ProbeShift: Predicting Probe-Direction Rotation from Stability is Largely a Sampling-Noise Floor.* **ACML 2026** CORE C, conference track — **Oral**, PMLR.

Linear probes read concepts such as truth from hidden states, but their direction drifts when inputs are reworded. A stability signal from in-distribution data alone seemed to predict this drift almost perfectly ($\rho = 0.945$), yet it predicted a placebo with no rewording even better ($\rho = 0.966$), so it mostly tracked sampling noise. Only one of seven signals kept a real effect, shown only for back-translated paraphrases.

**[3] K. Zhang**. *Grounding or Lexical Overlap? A Matched-Pair Study of Internals-Based Faithfulness Probes.* **GroundLM 2026** (workshop at EMNLP) — workshop **Findings** track.

Hallucination detectors that read a "faithful to the context" direction from hidden states may just spot copied words: in the standard extractive QA setup an answer is faithful exactly when its words appear in the context ($r = 0.99$). On matched pairs holding word overlap fixed, the learned direction mostly tracked whether an answer fits the question; on real hallucinations (RAGTruth, three models) it trailed NLI and word overlap.

**[4] K. Zhang**, C. Xia†. *A Point-in-Time Audit of Graph-Snapshot Leakage in GNN Stock Selection.* **PRICAI 2026** CORE B — regular paper (12–16 pp.), **Oral**, Springer LNAI.

Stock-selection GNNs are often backtested on a graph built at the end of the sample, which may include links that were not yet public. I rebuilt it using only fund holdings (SEC 13F) already public at each date. Two apparent leaks vanished with more seeds and random-graph controls; in a daily test the end-of-sample graph scored higher mainly by being more stable.

**[5] K. Zhang**. *PCP: A Controlled Benchmark for Pragmatic Intention Recognition in Clinical Communication.* **LUHME 2026** @ EMNLP — **Oral**.

In clinical talk people often hedge, play down serious worries or ask for help politely. On my 3,533-item benchmark, ten of eleven models picked what the speaker wanted 95.5–98.4% of the time but named the cue type only 56.6–64.5%; API models near ceiling on broad synthetic items fell to 79.7–90.6% on items adapted from public doctor–patient dialogues.

### UNDER REVIEW

Three further first-author manuscripts are under review, two of them now in Phase 2; titles and venues are withheld during anonymous review.

## Research Themes

- **LLM evaluation, alignment and post-training**: evaluation protocols and benchmarks [1] [5]; validity of probes on model internals [2] [3]; leakage and confounds in forecasting [4] [under review]; alignment and post-training [under review]

## Industry Experience

**Chinte Technology** Changsha, China
*AI Application Development (intern) — built a retrieval-augmented generation architecture* Nov 2024 – Jun 2025

**Guangdong Lianhe Information Technology** China
*Data Analysis (intern) — business data evaluation* Feb 2024 – Aug 2024

## Academic Service

**Reviewer** NLPCC 2026 (main conference) • GroundLM 2026 (workshop at EMNLP) • VLM4RWD 2026 (workshop at NeurIPS) 2026

**Student member** ACL • CCF (China Computer Federation)

## Awards

- 2025 **International Excellence Scholarship** (£8,438), School of Computer Science, University of Leeds.
- May 2025 **Inspiring Excellence Scholarship** (£10,000), Durham University.
- Apr 2024 **National First Prize**, 2nd "Waiwen Award" National College English Vocabulary Competition — **English Majors Division**. China Foreign Languages Publishing Administration.
  Awarded in the English Majors division as a Computer Science student.
- Jul 2024 **Second Prize (national)**, 7th National College Students' Arts Festival — Ensemble Performance, Group A. **Ministry of Education of the PRC**.
- 2023–24 **Second-Class Scholarship**, Hunan University of Finance and Economics.

## Competitions

**Kaggle Competitions Expert** — **1,957 of 217,591** (top 0.9%) • **Notebooks Expert** — **89 of 60,915** (top 0.1%)
Both medals won **solo**: **silver**, NVIDIA Nemotron Model Reasoning Challenge (**101 of 4,185**, top 2.4%); **bronze**, Orbit Wars (**409 of 4,729**, top 8.6%).

## Technical

**Frameworks** PyTorch, HuggingFace Transformers, scikit-learn, PyTorch Geometric

Referees and their contact details are available on request.